

DAMOS: The Smart Cruise Control for RAM

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TL; DR

- DAMOS is a Linux kernel feature for *automated* access-aware system operations



Image is retrieved from
https://pollackgroup.com/wp-content/uploads/2023/07/62bcd9b81b5cfd7bf26cb65d_self-driving-cars-1-jpg.webp

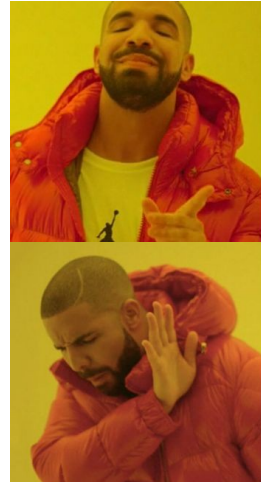
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- Why Smart Control for RAM (6 minutes)
- Begins: Origin to Productizing (10 minutes)
- Darknight: DAMOS Auto-tuning (10 minutes)
- Rises: DAMOS Usages (2 minutes)
- Revolution: Ongoing Changes (2 mins)
- QnA (10 mins)

Why Smart Control for RAM

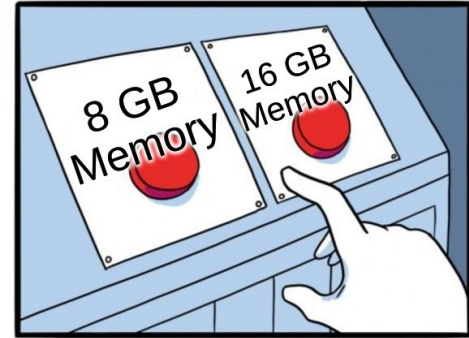
RAM Efficiency Matters

- Data: Source of money
 - Getting bigger :D
- RAM (Memory): Sink of money
 - Getting more expensive :'(
- Goal: More data with less memory



Real World Story

- In a cloud provider far, far away
- CPU: Pay-as-you-go
- Memory: Fixed price
- Plan: pay-as-you-go for memory
 - Charge for only working set

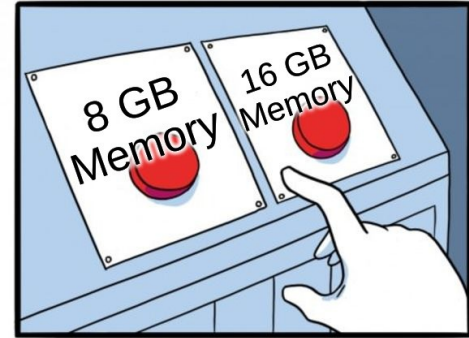


imgflip.com

JAKE-CLARK.TUMBLR

Real World Story

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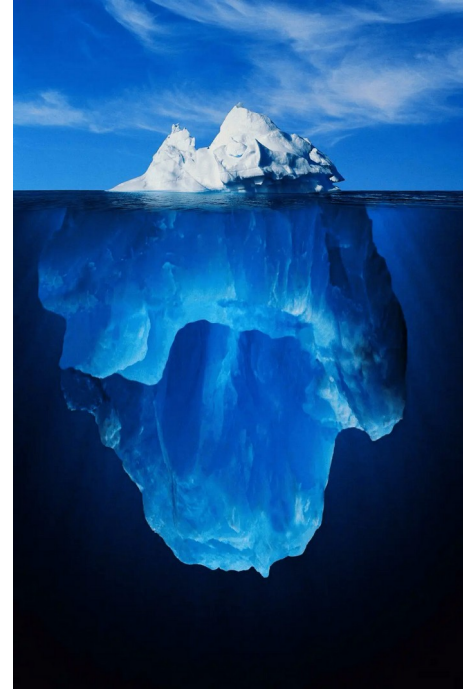


“Everyone Has a Plan Until They Get Punched in the Mouth”,
Mike Tyson

Image is retrieved from <https://imgflip.com/i/b1mr1j>

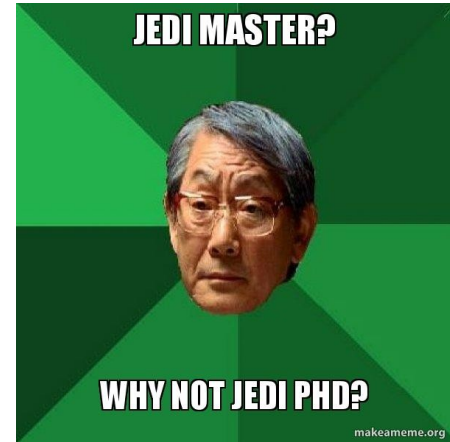
Problem: Working Set Detection

- Working set $\neq$ Allocated memory
- Much more difficult than it looks like
 - Definition of working set
 - Monitoring overhead
 - Memory-greedy system
- Common and essential for memory efficiency



DAMON: A New Hope

- A PhD student's side project
 - For the issue in main research
 - Developed a kernel subsystem as a solution
- Named DAMON: Data Access MONitor



DAMON in Nutshell

DAMON in One Sentence

“DAMON is a Linux kernel subsystem for efficient access monitoring and access-aware system operations.”

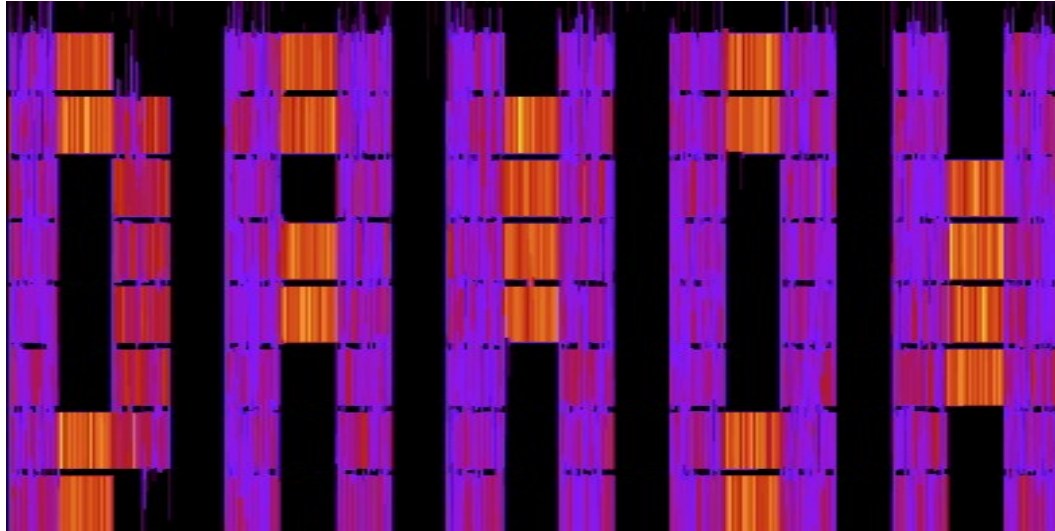


Image is [generated](#) by DAMON

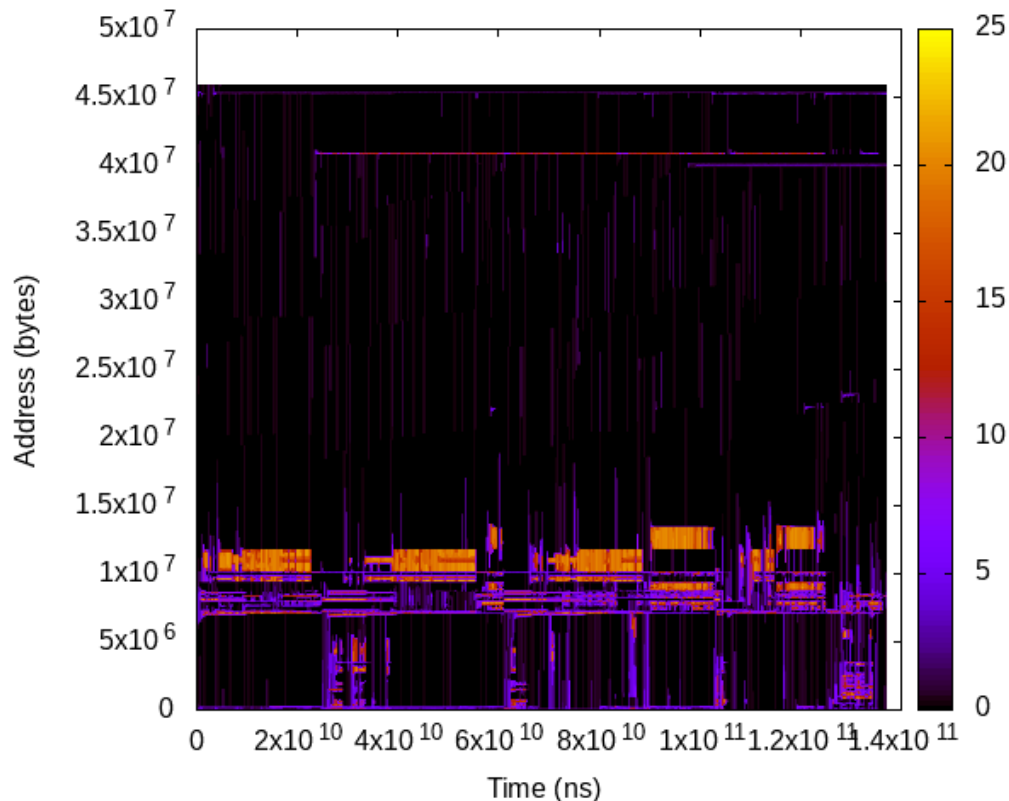
High Level Implementation

- Spawn a kernel thread that does below
- See if each page is accessed, every 5ms
- Inform users the findings, every 100ms
- Time intervals can be [auto-]tuned

```
$ ls mm/damon/  
acma.c          Makefile          ops-common.h      sysfs.c           sysfs-schemes.c  
core.c          modules-common.c  paddr.c          sysfs-common.c    tests  
Kconfig        modules-common.h  reclaim.c        sysfs-common.h    vaddr.c  
lru_sort.c      ops-common.c      stat.c           sysfs-sample.c  
$ wc -l mm/damon/*.c | tail -n 1  
16000 total
```

Providing Access Information

- Location
- Frequency
- Stability
- Recency



Access pattern heatmap of
Splash2x/Raytrace

Design

- Goals: Accurate, Lightweight, Tunable and Automated
- Mechanisms: Sampling, Adaptive regions, Sysfs interface, Feedback loops
- Results
 - $<0.1\%$ CPU overhead on products

Availability and Usages

- Available on ≥ 5.15 upstream kernels
- Enabled on most [distro](#) kernels
- Being used for products and researches

Community

- Public channels
 - Mailing list: damon@lists.linux.dev
 - Project website: <https://damonitor.github.io/>
- Private channels
 - Maintainer email: sj@kernel.org
 - DAMON Beer/Coffee/Tea [Meetup](#)

But This is Not About DAMON

- DAMON was [presented](#) in detail at Kernel Recipes'25



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DAMOS Begins

Developer's Expectation

- DAMON only provides Access Pattern
- Users do hard lifts
 - Decide optimum system memory size
 - Optimize programs for access pattern



Image is retrieved from <https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSOuahkkiJvsAe56zRaoB0KSAqMdwUS5XB36ULDvioD0HGmRe2d4GJYILc&s=10>

DAMON User Reaction in Real

- Users are busy, even in hyper-scalers
- No time to run a tool for no direct wins
- FAQ: Can DAMON provide wins itself?

What updating my status felt like



The reality



Image is retrieved from [https://encrypted-tbn0.gstatic.com/images?](https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSOuahkkiJvsAe56zRaoB0KSAqMdwUS5XB36ULDvioD0HGmRe2d4GJYILc&s=10)

[q=tbn:ANd9GcSOuahkkiJvsAe56zRaoB0KSAqMdwUS5XB36ULDvioD0HGmRe2d4GJYILc&s=10](https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSOuahkkiJvsAe56zRaoB0KSAqMdwUS5XB36ULDvioD0HGmRe2d4GJYILc&s=10)

madvise()

- System call
- Gives memory advice to kernel
- Supported actions
 - MADV_PAGEOUT,
MADV_HUGEPAGE,
MADV_WILLNEED, ...



DAMON+madvise()

- Let DAMON do `madvise()` calls
- For memory of a given pattern
 - Pattern: Access frequency & stability
- Target pattern+`madvise()` action:
DAMon-based Operation Scheme
(DAMOS)

Still a better effort than
"Thoughts And Prayers"



Image is retrieved from

https://static0.hotcarsimages.com/wordpress/wp-content/uploads/2018/05/6138666188_f86c3cdc67_b.jpg?q=50&fit=crop&w=825&dpr=1.5

DAMON+madvise() Example

- Proactive cold memory reclaim
 - ‘damo’ is DAMON user-space **tool**

```
$ sudo ./damo start $(pidof my_workload) \  
    --damos_action pageout \  
    --damos_access_rate 0% 0% --damos_age 2m max
```

DAMON+madvise() Example

- Proactive cold memory reclaim

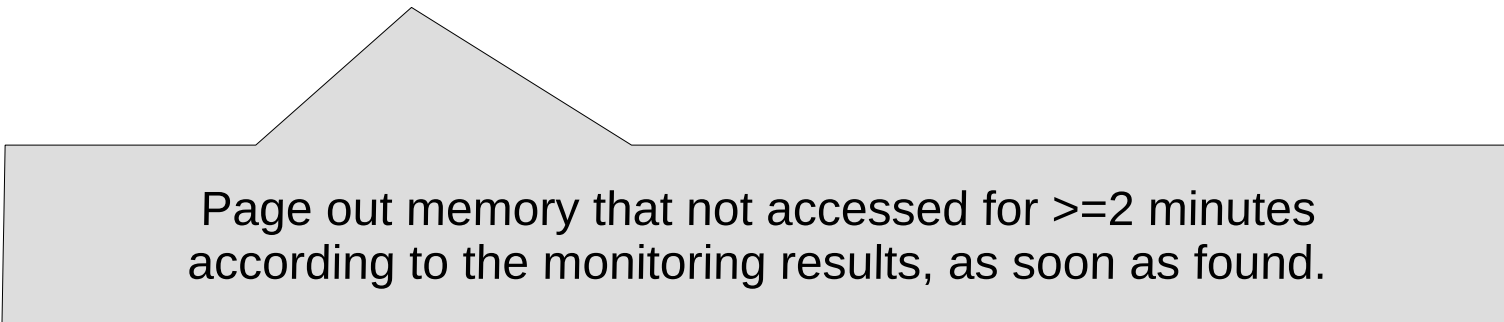
```
$ sudo ./damo start $(pidof my_workload) \  
--damos_action pageout \  
--damos_access_rate 0% 0% --damos_age 2m max
```

Start DAMON, monitor access pattern of 'my_workload' process

DAMON+madvise() Example

- Proactive cold memory reclaim

```
$ sudo ./damo start $(pidof my_workload) \  
    --damos_action pageout \  
    --damos_access_rate 0% 0% --damos_age 2m max
```



Page out memory that not accessed for ≥ 2 minutes according to the monitoring results, as soon as found.

More than `madvise()`

- No reason to support only `madvise()` actions
- DAMOS is later extended for general system operations
 - Moving folio inactive → active LRU list
 - Migrate folio to a given node

DAMOS for Products

What If DAMOS Out Of Control

- Access pattern is dynamic
 - Target memory can suddenly increase
 - DAMOS can consume huge system resource
 - Critical in real-world products



Image is retrieved from

https://inews.co.uk/wp-content/uploads/2022/01/JGD_1214107.jpg?w=1200&h=675&crop=1

DAMOS Quota

- Max speed of DAMOS actions
 - E.g., reclaim cold memory up to 128 MiB/s
- Regions for action under quota
 - Selected via access pattern
 - Hottest or coldest memory, depend on action



Image is retrieved from

<https://i.pinimg.com/736x/ba/31/88/ba3188f5a6f7a1b6227262a14d9841ae.jpg>

Special Requests

- Special products have unique requests
 - Access pattern is useful but not everything
 - Example requests
 - Reclaim cold page cache but anon pages
 - Reclaim cold memory of cgroup X but Y

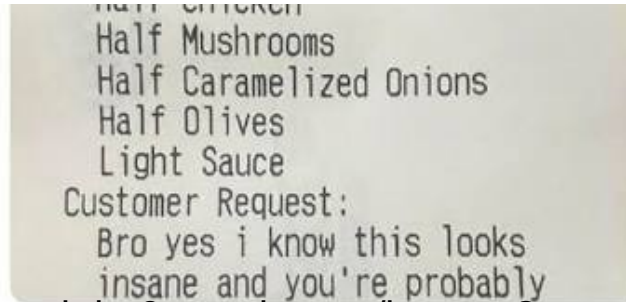


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[q=tbn:ANd9GcRuwoXpDo3YjK9u8JQ8r0VVP5YWcNENt9NaxBS2uicKEH4nG-ef_YAIUo4&s=10](https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRuwoXpDo3YjK9u8JQ8r0VVP5YWcNENt9NaxBS2uicKEH4nG-ef_YAIUo4&s=10)

DAMOS Filter

- Non-access pattern target spec
- Allow/reject memory of given types
- Support multiple filters combination
 - “Reclaim anon pages of cgroup A”



DAMOS Quota, Filters Example

- Speed-bounded, Targeted proactive cold memory reclaim

```
$ sudo ./damo start \  
    --damos_action pageout \  
    --damos_access_rate 0% 0% --damos_age 2m max \  
    --damos_quota_space 128M --damos_quota_interval 1s \  
    --damos_filter reject anon \  
    --damos_filter reject memcg /workload_latency_sensitive
```

DAMOS Quota, Filters Example

- Speed-bounded, Targeted proactive cold memory reclaim

```
$ sudo ./damo start \  
--damos_action pageout \  
--damos_age_min 0% 0% --damos_age 2m max \  
--damos_quota 128M --damos_quota_interval 1s \  
--damos_filters
```

Start DAMON,
monitor access pattern of entire physical address space.

ive

DAMOS Quota, Filters Example

- Speed-bounded, Targeted proactive cold memory reclaim

```
$ sudo ./damo start \  
  --damos_action pageout \  
  --damos_access_rate 0% 0% --damos_age 2m max \  
  --damos_quota_space 128M --damos_quota_interval 1s \  
  --damos_filter reject anon \  
  --damos_filter reject memcg /workload_latency_sensitive
```

Pageout memory that not accessed for ≥ 2 minutes according to the monitoring output, as soon as found.

DAMOS Quota, Filters Example

- Speed-bounded, Targeted proactive cold memory reclaim

```
$ sudo ./damo start \  
  --damos_action pageout \  
  --damos_access_rate 0% 0% --damos_age 2m max \  
  --damos_quota_space 128M --damos_quota_interval 1s \  
  --damos_filter reject anon \  
  --damos_filter reject memcg /workload_latency_sensitive
```

However, limit the pageout to be tried only up to 128 MiB per second.
If there are more than 128 MiB cold memory,
do reclaim coldest 128 MiB memory.

DAMOS Quota, Filters Example

- Speed-bounded, Targeted proactive cold memory reclaim

\$ sudo .

Also, skip the reclaim for anonymous pages.

```
--damos_age 0% 0% --damos_age 2m max \  
--damos_quota_space 128M --damos_quota_interval 1s \  
--damos_filter reject anon \  
--damos_filter reject memcg /workload_latency_sensitive
```

DAMOS Quota, Filters Example

- Speed-bounded, Targeted proactive cold memory reclaim

\$ sudo .

Skip the reclaim for memory that belongs to
'workload_latency_sensitive' memory cgroup, too.

```
--damos_quota 128M --damos_quota_interval 1s \  
--damos_filter reject anon \  
--damos_filter reject memcg /workload_latency_sensitive
```

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Dark Knight: DAMOS Auto-tuning

Problem: Tuning Difficulties

- DAMOS users should...
 - Understand DAMON/DAMOS internal
 - Understand their workload access pattern
 - Setup optimum scheme for the workload
- Complicated, repetitive for each workload

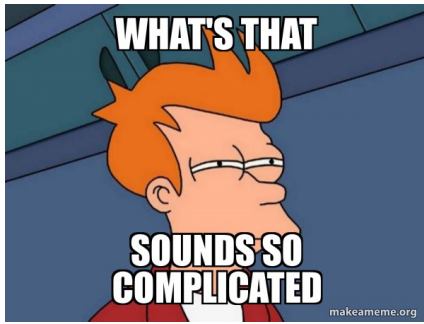


Image is retrieved from
<https://media.makeameme.org/created/whats-that-sounds.jpg>

Why So Difficult? It's Question!

- DAMOS asks users “how” (mechanism)
- Effectively programming request
- Flexible but difficult



Image is retrieved from

<https://i.pinimg.com/736x/4c/25/6f/4c256f103aca4785544b78e430c079aa.jpg>

Change The Question

- Idea: Ask users “what” (policy, or *vibe*)
 - What they want to achieve
 - E.g., 5% free memory, 0.05% memory pressure
 - Intuitive for service owners
- DAMOS works in darkness
 - Auto-tune parameters to achieve the goal

Knobs to tune: Access Pattern?

- Min/max access frequency and stability
- Four parameters; too complicated



Knobs to Tune: Quota!

- Single parameter; much simple
- Effectively auto-tune access pattern parameters



Image is retrieved from

https://thumbs.static-thomann.de/thumb/padthumb600x600/pics/bdb/_28/284224/10785205_800.jpg

DAMOS quota auto-tuning

- Users set goals of the quota
 - Goal: a target value of a system metric
- Auto-tune quota using a feedback loop
 - Increase/decrease quota when the goal is under/over-achieved
 - Multiple tuning algorithms available

Auto-tuned DAMOS Example

```
$ sudo ./damo start \  
  --damos_action pageout \  
  --damos_access_rate 0% 0% --damos_age 2m max \  
  --damos_quota_space 128M --damos_quota_interval 1s \  
  --damos_filter reject anon \  
  --damos_filter reject memcg /workload_latency_sensitive  
  --damos_quota_goal some_mem_psi_us 5ms
```

Auto-tuned DAMOS Example

```
$ sudo ./damo start \  
  --damos_action pageout \  
  --damos_access_rate 0% 0% --damos_age 2m max \  
  --damos_quota_space 128M --damos_quota_interval 1s \  
  --damos_filter reject anon \  
  --damos_filter reject memcg /workload_latency_sensitive  
  --damos_quota_goal some_mem_psi_us 5ms
```

Start DAMON, monitor entire physical memory, reclaim memory that
not accessed for ≥ 2 minutes, up to 128 MiB/s,
skipping anonymous pages and pages of latency sensitive workload
cgroup

Auto-tuned DAMOS Example

```
$ sudo ./damo start \
```

Tune the quota to achieve 5ms/1s (0.05%) memory pressure stall.
If <0.05% memory pressure, reclaim more cold page, and vice versa.
Little pressure is good for health :)

```
--damos_fi reject memcg /workload_latency_sensitive  
--damos_quota_goal some_mem_psi_us 5ms
```

DAMOS for Hot Memory

- DAMOS is not only for cold memory
- Memory tiering: one of DAMOS use cases



Image is retrieved from

<https://i.pinimg.com/736x/51/8f/09/518f09467228e2ac87c0f2d299d02fc5.jpg>

Memory Tiering 101

- Two tiers: DRAM and CXL memory
 - DRAM: Fast, small
 - CXL: Slow, large
- Goal: Hot data on DRAM, cold data on CXL

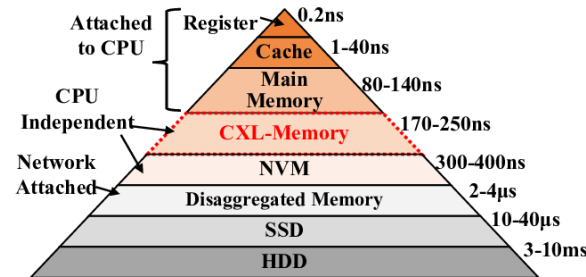


Image is retrieved from

<https://www.researchgate.net/publication/361162182/figure/fig1/AS:1164536778358786@1654658612492/Different-tiers-has-different-characteristics-with-varied-memory-access-latency-in-a.png>

Transparent Page Placement

- Utilize DRAM as much as possible
 - It is more powerful, no reason to under utilize
- Keep headroom free memory in DRAM
- TPP-DAMON: DAMOS-based TPP implementation

TPP-DAMON

```
damo start \  
  --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_cold 1 --damos_access_rate 0% 0% \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_free_bp 0.5% 0 \  
    --damos_filter reject young \  
  --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_hot 0 --damos_access_rate 5% max \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_used_bp 99.7% 0 \  
    --damos_filter allow young \  
    --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \  
  --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

TPP-DAMON

damo start \

Dear DAMON user-
space tool,

Start DAMON in
kernel.

```
--numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
--damos_action migrate_hot 0 --damos_access_rate 0% 0% \
--damos_apply_interval 1s \
--damos_quota_interval 1s --damos_quota_space 200MB \
--damos_quota_goal node_mem_free_bp 0.5% 0 \
--damos_filter reject young \
--numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \
--damos_action migrate_hot 0 --damos_access_rate 5% max \
--damos_apply_interval 1s \
--damos_quota_interval 1s --damos_quota_space 200MB \
--damos_quota_goal node_mem_used_bp 99.7% 0 \
--damos_filter allow young \
--damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
--nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

TPP-DAMON

```
damo start \
```

```
--numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
```

```
--damos_action migrate_cold 1 --damos_access_rate 0% 0% \
```

```
--damos_apply_interval 1s \
```

Create one DAMON
worker thread, monitoring
for node 0 (upper tier)

```
--damos_quota_interval 1s \
```

```
--damos_quota_goal node_mem_used_bp 99.7% 0 \
```

```
--damos_filter allow young \
```

With auto-tuned
monitoring intervals.

```
200MB \
```

```
--numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \
```

```
--damos_action migrate_hot 0 --damos_access_rate 5% max \
```

```
--damos_apply_interval 1s \
```

```
--damos_quota_interval 1s --damos_quota_space 200MB \
```

```
--damos_quota_goal node_mem_used_bp 99.7% 0 \
```

```
--damos_filter allow young \
```

```
--damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
```

```
--nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

TPP-DAMON

```
damo start \  
  --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_cold 1 --damos_access_rate 0% 0% \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_used_bp 99.7% 0 \  
    --damos_filter allow young \  
    --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \  
  --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

And ask it to *migrate* un-accessed (access_rate 0%)
memory to node 1 (lower tier), *coldest* one first.
IOW, **demote** cold memory.

TPP-DAMON

```
damo start \  
  --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_cold 1 --damos_access_rate 0% 0% \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_used_bp 99.7% 0 \  
    --damos_filter reject young \  
  --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_hot 0 --damos_access_rate 5% max \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_used_bp 99.7% 0 \  
    --damos_filter allow young \  
    --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \  
  --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

And do that once per second.

TPP-DAMON

```
damo start \  
  --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_cold 1 --damos_access_rate 0% 0% \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_free_bp 0.5% 0 \  
    --damos_filter allow young \  
  --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_cold 1 --damos_access_rate 0% 0% \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_used_bp 99.7% 0 \  
    --damos_filter allow young \  
  --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \  
  --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Within *one second*, do the migration only up to *200 MiB* of memory.

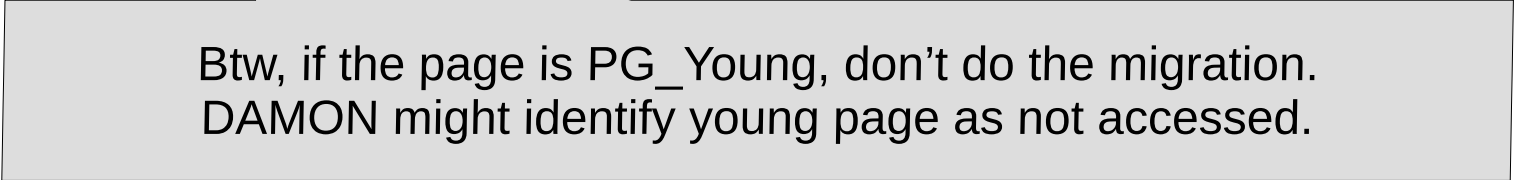
TPP-DAMON

```
damo start \  
  --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_cold 1 --damos_access_rate 0% 0% \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_free_bp 0.5% 0 \  
    --damos_filter reject young \  
  --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_cold 1 --damos_access_rate 0% 0% \  
    --damos_apply_interval 1s \  
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    --damos_filter allow young \  
  --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \  
--nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Under the 200MiB/1s hard limitation, auto-tune the real soft limitation to make node 0's free memory ratio becomes 0.5% soon.

TPP-DAMON

```
damo start \  
  --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_action migrate_cold 1 --damos_access_rate 0% 0% \  
    --damos_apply_interval 1s \  
    --damos_quota_interval 1s --damos_quota_space 200MB \  
    --damos_quota_goal node_mem_free_bp 0.5% 0 \  
    --damos_filter reject young \  
  --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \  
    --damos_access_rate_hot 0 --damos_access_rate_cold 5% max \  
    --damos_filter allow young \  
    --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \  
  --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```



Btw, if the page is PG_Young, don't do the migration.
DAMON might identify young page as not accessed.

TP

damo

Create another DAMON worker thread that monitors *node 1* (lower tier) with auto-tuned monitoring intervals.

Ask it to migrate memory shown $\geq 5\%$ access rate to *node 0* (upper tier), *hottest* first. IOW, **promote** hot pages.

Do the migration once per second, up to 200 MiB of memory.

Under the *200 MiB/s* hard limit, adjust the real soft limit aiming *node 0* memory utilization ratio 99.7%.

Btw, to that only if the page is *PG_Young*.

```
--damos_... node_mem_tree_bp 0.5% 0 \
--damos_filter reject young \
--numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \
--damos_action migrate_hot 0 --damos_access_rate 5% max \
--damos_apply_interval 1s \
--damos_quota_interval 1s --damos_quota_space 200MB \
--damos_quota_goal node_mem_used_bp 99.7% 0 \
--damos_filter allow young \
--damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
--nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

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Dark Knight Rises: Publicly Known DAMOS Usages

Proactive Memory Reclaim

- AWS Aurora Serverless v2
- Proactively reclaim cold memory in VMs
- Pay for only remaining non-cold memory



DRAM/CXL Memory Tiering

- SK Hynix **HMSDK**
 - Migrate hot/cold data across tiers
- Micron dynamic **interleaving**
 - Interleave hot/cold memory for bandwidth



More Non-public Usages

- Real DAMOS usages are unknown
- A few more usages are privately shared
- Make your own creative usage

Ongoing DAMON/S Revolution

The Classes

- Primary class: Access information
- Secondary class: Data attributes
 - DAMOS filters for page level properties

Is It Fair (and Useful)?

- Why access info is always primary?
- Other attributes are primary interests for some users

DAMON Revolution

- Re-define DAMON
 - Data Attributes Monitoring/Operation eNgin
- Every attribute including access is equal
- DAMOS get revolutionized together



Image is retrieved from <https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcShaoxj5LAQw3g2Ed4w83aLruDWdILAN1TRojKmw49EKQ&s=10>

Revolution In Progress

- Monitoring framework is extended
- More support of attributes is ongoing
 - Now: access, Anon page, memcg, hugepage
- Minimum support in DAMOS yet

Revolution Never Ends

- DAMON evolves for users

Wrapup

- DAMOS is a Linux kernel feature for *automated* access-aware system operations
- Being used for products
- A revolution for making it for every attribute is ongoing

QnA

- Don't forget
 - sj@kernel.org
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 - <https://damonitor.github.io>
 - Just poke me in hallway